

AI Deep Drawability

August 2026 Project Update

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1 Brief Update

We previously identified that Sheets 4 and 2123 seem to be outliers and have some underlying difference in structure that separate them from the rest of their class. If we drop these from our dataset we see a substantial improvement in model performance, going from an average accuracy of **0.635** with a standard deviation of 0.164 to an accuracy of **0.753** with a standard deviation of just 0.043. Figure 1, shows the spread of accuracies for each holdout combo side by side. We can see that after dropping sheets 4 and 2123 our lowest accuracy holdout is around 0.67, significantly better than the previous results under 0.3 with sheet 2123.

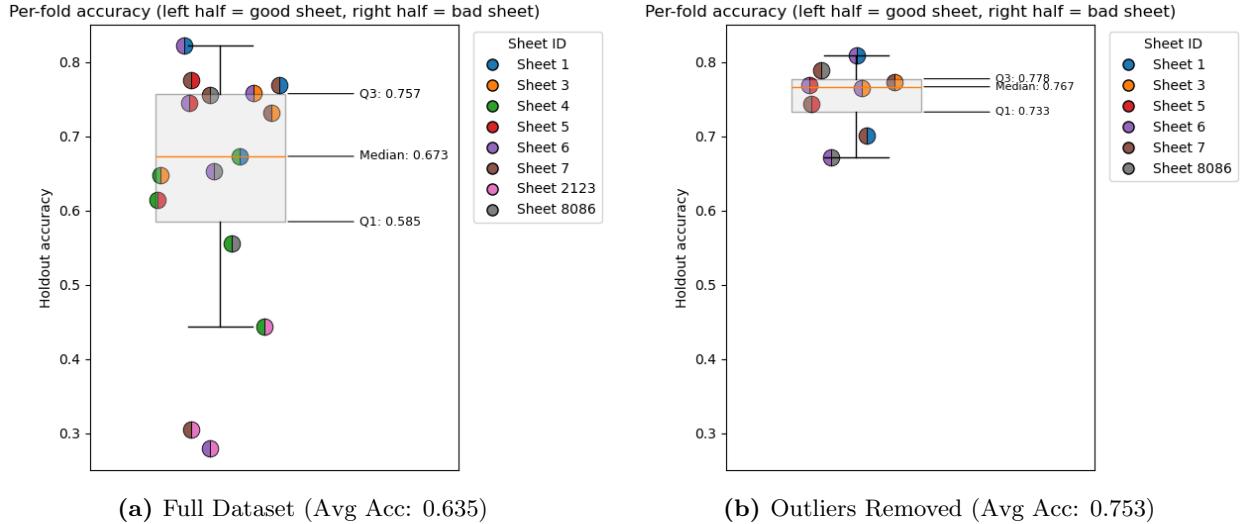


Figure 1: Per holdout accuracies before and after dropping outlier sheets.

Figure 2 shows these per-holdout accuracies in a numerical format, listing each accuracy to 3 decimal places. We can see that holding out Sheets 1 and 6 we get our best holdout accuracy of 0.809 and holding out sheets 6 and 8086 we get our lowest accuracy of 0.672. Interestingly, the best here is actually slightly lower than we saw when testing with the full data of 0.823 but the low is substantially higher than the 0.279 we had previously. This indicates that we might be able to squeeze some more performance out of the model by tuning for this reduced dataset since we are still using the hyperparameters that we found were best for the full dataset. But even with this extra tuning our model will not generalize better to truly different new samples like 2123 and 4. The bigger issue is that we still have too few unique samples.

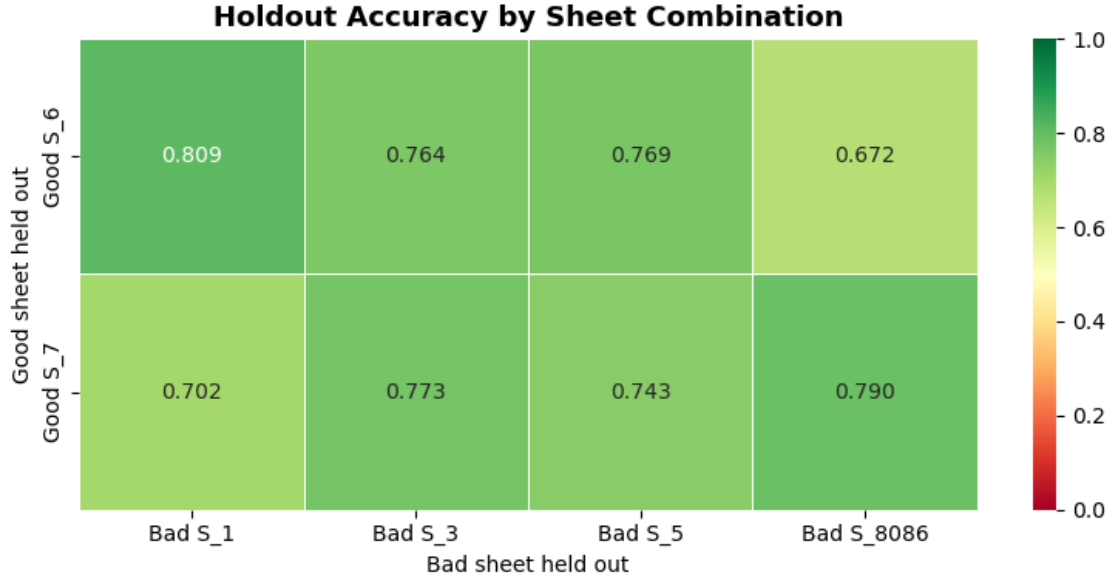


Figure 2: Accuracy (w/out sheets 4, 2123)

In Figure 3 (next page) we see the updated confusion matrix with those outliers removed. As we can see, the majority vote for each held out sheet now is for a sheet of the same class, as we would hope. Focusing on the bad sheets first (1, 3, 5, and 8086) we can see that even the second and third highest probability sheets are also of the same class and the probabilities of it being from the opposite class are far least likely. This gives us a good indication that our model is learning similarities between different bad sheets. Since we only have two good sheets and one is always held out there is only one choice of the same class and for both 6 and 7 we see this choice is selected. But this means that the second and third votes for will be for the opposite class simply because there are no other good samples to choose from.

Overall, these results are promising and seem to indicate that the majority of our unique sheets do share common characteristics with the other sheets of their class.

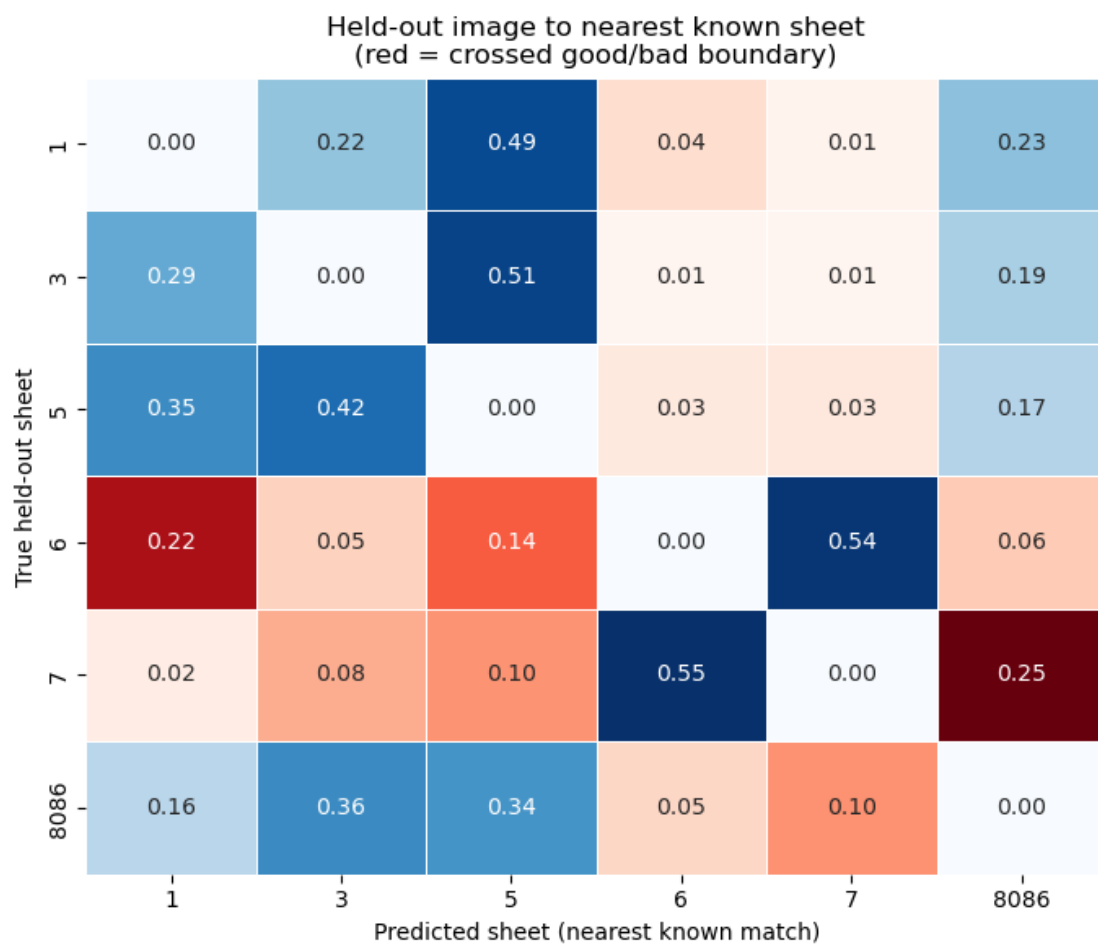


Figure 3: Confusion matrix (w/out sheets 4, 2123)